Discrete Probability Theory (IN0018)
School of Computation, Information, and Technology Technical
University of Munich

Tutorial 00

Set Theory & σ-Algebras

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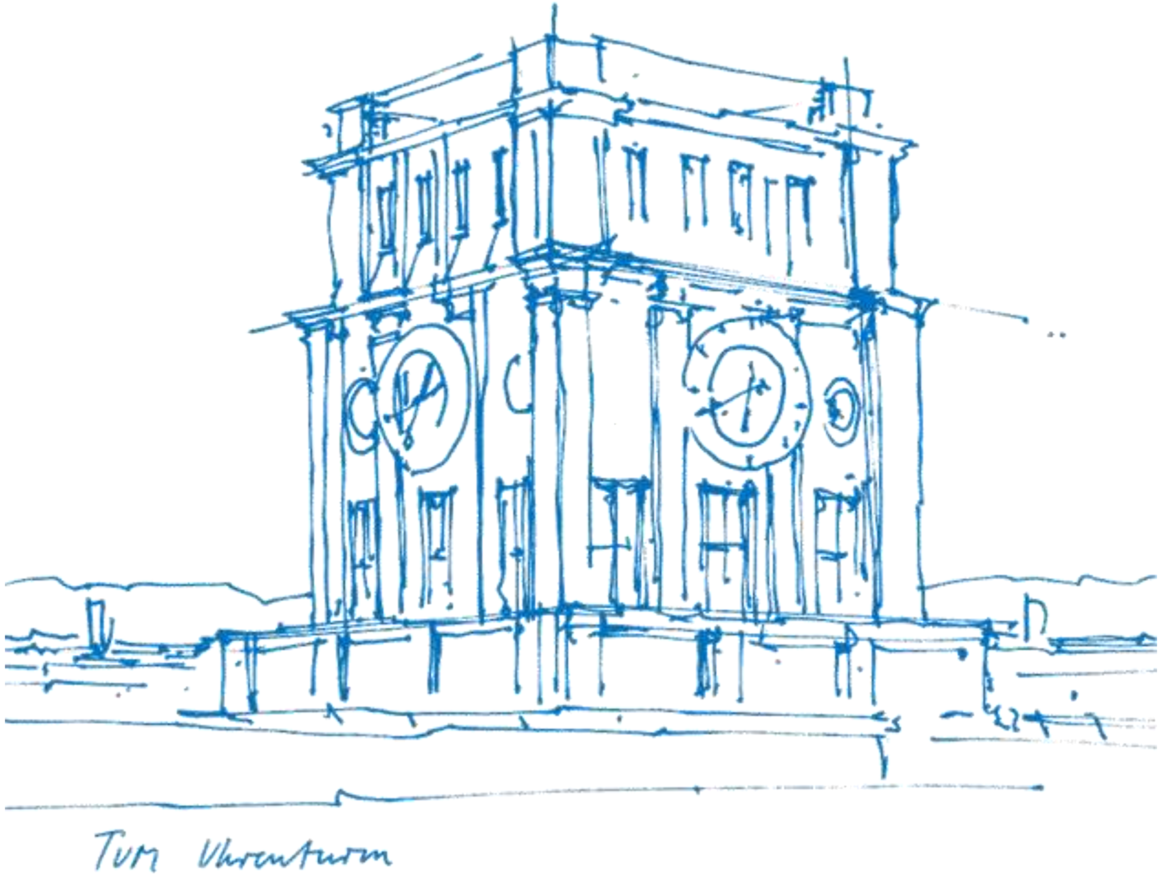
20. April / 21. April 2026



Intro

- **Lecture:** 2.5h per week at 16:00-18:30 Wed, held by Porf. Niki Kilbertus in room MW2001

- **Tutorial:** In-Person, 1.5h by two time periods per week - 12:15-13:45 Mon, 00.08.053 (A31)
  - 14:15-15:45 Tue, 00.08.053 (B42)

- The slides are not guaranteed to be accurate! (Written by myself)

- **Language during the tutorial:** Mainly English, Deutsch may be used if needed, Chinese might be used during calculation (that would be an exception and will be repeated in English)

- **Exam:** August 5, 2026, 08:00-10:00, Cheatsheet allowed, Calculator banned

- **Materials of this tutorial:** Please scan the QR-Code ☞

- **Contact:** xujie.zhao@tum.de

Overview (referring to last year's lecture)

- **Discrete Probability**
  - Probability Space, Event, Random Variable
  - Special Distributions
  - Markov's and Chebyshev's Inequalities
- **Continuous Probability**
  - Normal Distribution, Exponential Distribution
  - Central Limit Theorem
- **Statistics**
  - Estimators
  - Confidence Intervals
  - Hypothesis Testing

△ Derivative and Integral required!
(Analysis für Informatiker MA0902)

Personal advice...

Back to us!

- PART I. Recap of the Lecture

- PART II. Exercises from the Exercise Sheet

- PART III. Additional Exercises (with solution)

I. Recap

- **Set Theory**
  - Basic Definitions
  - De Morgan's Laws
  - Samples (sample space), Events, Event Systems

- **σ-Algebra**
  - Definition
  - Intersection of σ-algebras
  - Generator of a σ-algebra

- **Measurable space and sets**
  - Definition
  - Induced σ-algebra

- **Measure and measure space**
  - Definition
  - Properties of measures

II. Exercise Sheet

Exercise 1. Let $\Omega \neq \emptyset$ and let $\{A_\alpha \in \mathcal{P}(\Omega) \mid \alpha \in I\}$ be an arbitrary set of subsets of Ω, I the indexing set. Then the following two identities hold (commonly referred to as *de Morgan's law*):

$$\left(\bigcap_\alpha A_\alpha\right)^{\mathsf{c}} = \bigcup_\alpha A_\alpha^{\mathsf{c}}, \qquad \left(\bigcup_\alpha A_\alpha\right)^{\mathsf{c}} = \bigcap_\alpha A_\alpha^{\mathsf{c}}$$

II. Exercise Sheet

Exercise 2. Let Ω_1, Ω_2 be non-empty sets, $f : \Omega_1 \to \Omega_2$ an arbitrary mapping and $\mathcal{C} \subseteq \mathcal{P}(\Omega_2)$ an arbitrary collection of subsets of Ω_2. Then the following statements hold.

(i) If $A, B \subseteq \Omega_2$ and $A \subseteq B$, then $f^{-1}(A) \subseteq f^{-1}(B)$.

(ii) The inverse image of the union is equal to the union of the inverse images, meaning

$$f^{-1}\Big(\bigcup_{A\in\mathcal{C}} A\Big) = \bigcup_{A\in\mathcal{C}} f^{-1}(A) \ .$$

(iii) Given a subset $A \subseteq \Omega_2$, the inverse-image of the complement is equal to the complement of the inverse image, meaning

$$f^{-1}(A^{\mathsf{c}}) = \left(f^{-1}(A)\right)^{\mathsf{c}}$$

II. Exercise Sheet

Exercise 3. Suppose $\mathcal{A}$ is a σ-algebra over $\Omega \neq \emptyset$, let $A_0, A_1, A_2, \ldots$ be a countable collection of sets in $\mathcal{A}$ and let $m \in \mathbb{N}$. Then the sets

$$\Omega, \quad A_0 \setminus A_1, \quad \bigcup_{i=0}^{m} A_i, \quad \bigcap_{i=0}^{m} A_i, \quad \bigcap_{i \in \mathbb{N}} A_i$$

are also in $\mathcal{A}$.
[Hint: Use de Morgan's law for the intersection properties.]

II. Exercise Sheet

Exercise 4. True or false? Justify your answers either by a proof or a counterexample.

(i) The set $\mathcal{A} = \{\emptyset, \{1,2\}, \{1,4\}, \{2,3\}, \{3,4\}, \{1,2,3,4\}\} \subseteq \mathcal{P}(\Omega)$ is a σ-algebra over $\Omega = \{1,2,3,4\}$.

(ii) If $\mathcal{A}_1, \mathcal{A}_2 \subseteq \mathcal{P}(\mathcal{A})$ are σ-algebras over Ω, then so is $\mathcal{A}_1 \cup \mathcal{A}_2$.

(iii) The set $\mathcal{A} := \{A \subseteq \Omega \mid A \text{ or } A^c \text{ is countable}\}$ is a σ-algebra over Ω.

(iv) The set $\mathcal{A} := \{A \subseteq \Omega \mid A \text{ or } A^c \text{ is finite}\}$ is a σ-algebra over Ω if and only if Ω is finite.

[Hint: You can use the following facts without further justification: The countable union of countable sets is again countable. Subsets of countable sets are again countable.]

II. Exercise Sheet

Exercise 5. Let $\Omega \neq \emptyset$ be a set, I be a nonempty index set (finite, countable or even uncountable) and let $\mathcal{A}_\alpha$ be a σ-algebra over Ω for each $\alpha \in I$ in the index set. Then $\mathcal{A} := \bigcap_{\alpha \in I} \mathcal{A}_\alpha$, the intersection of all the sigma-algebras is also a σ-algebra over Ω.

II. Exercise Sheet

Exercise 6. Let Ω_1, Ω_2 be nonempty sets, $f : \Omega_1 \to \Omega_2$ an arbitrary mapping and let $\mathcal{A}, \mathcal{B}$ be σ-algebras over Ω_1 and Ω_2 respectively. Then the *inverse image of $\mathcal{B}$ under f* and the *image or push-forward of $\mathcal{A}$ under f* defined as

$$f^{-1}(\mathcal{B}) := \{f^{-1}(B) \mid B \in \mathcal{B}\} \quad \text{and} \quad f_*(\mathcal{A}) := \{B \subset \Omega_2 \mid f^{-1}(B) \in \mathcal{A}\}$$

are σ-algebras over Ω_1 and Ω_2 respectively.
[Hint: Use statements from the second exercise.]

III. Additional Exersices

Exercise 7. Anna flips a fair coin countable but infinitely many times. The set Ω consists of all infinite sequences of the outcomes "Heads" and "Tails".

(a) Show that the set Ω is uncountable.

(b) Let $\mathcal{A}$ be the set of all subsets $A \subseteq \Omega$ such that either A and $\Omega \setminus A$ is countable. Prove that $\mathcal{A}$ is a σ-Algebra.

(a) Assume every natural number "i" maps to a sequence of coins with i heads, the sequence with all tails is then not included, therefore Ω uncountable.

(b) $\mathcal{A} = \{A \subseteq \Omega \mid A$ or $\bar{A}$ countable$\}$.

Rule 1. $\Omega \in \mathcal{A}$, since $\Omega \subseteq \Omega$ and $\bar{\Omega} = \phi$ countable.

Rule 2. $A \in \mathcal{A}$, A or $\bar{A}$ countable $\Rightarrow \bar{A} \in \mathcal{A}$, since $\bar{A} \subseteq \Omega$.

Rule 3. ① if every A_n countable, $\bigcup_{n=1}^{\infty} A_n$ countable, $\bigcup_{n=1}^{\infty} A_n \in \mathcal{A}$.

② if exists an uncountable A_i, $\bar{A_i} = \Omega \setminus A_i$ must be countable, $\overline{\left(\bigcup_{n=1}^{\infty} A_n\right)} = \Omega \setminus \bigcup_{n=1}^{\infty} A_n \in \Omega \setminus A_i$ countable.

$\Rightarrow$ For $\forall n$, $A_n \in \mathcal{A}$, then $\bigcup_{n=1}^{\infty} A_n \in \mathcal{A}$.

Materials